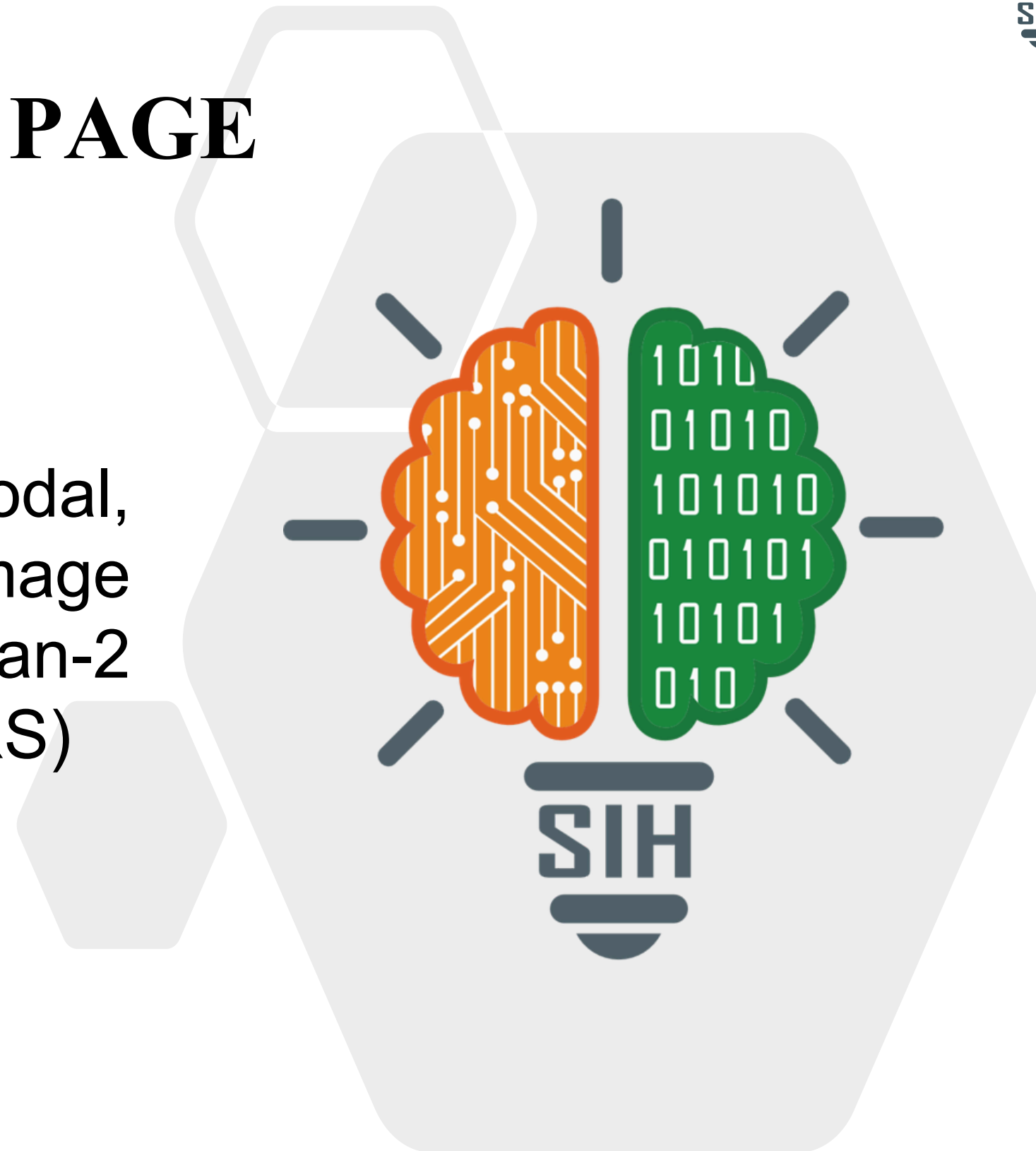


TITLE PAGE

- **Problem Statement ID** – 26166
- **Problem Statement Title**- Multi-modal, Sun Angle and Scale Invariant Image Correspondence using Chandrayaan-2 Optical Images (OHRC, TMC-2, IIRS)
- **Theme**- Space Technology
- **PS Category**- Software
- **Team ID**- SIH26-A0H-T109
- **Team Name**- HackRhythm



Multi-modal, Sun Angle and Scale Invariant Image Correspondence using Chandrayaan-2 Optical Images (OHRC,TMC-2,IIRS)



⚠ The Problem, in Plain Language

- **Extreme Scale Gap (320x)**

Severe spatial mismatch between **OHRC (0.25m)**, **TMC-2 (5m)**, and **IIRS (80m)** causes standard feature detectors (**SIFT/ORB**) to fail entirely.

- **Cross-Spectral Mismatch**

IIRS records mineral/spectral composition rather than reflected brightness, so pixel intensities aren't directly comparable to OHRC/TMC-2—a core reason simple pixel-matching fails across these payloads.

- **Dynamic Solar Illumination**

Low solar elevations (**15°–85° incidence**) in polar craters cause elongated, inverted shadow profiles that distort true crater rims.

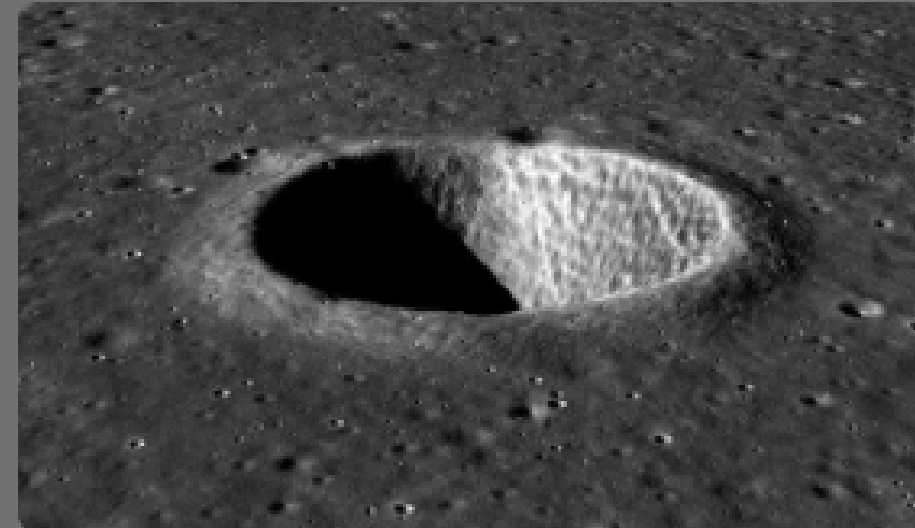
- **Geodetic Drift**

positioning and pointing uncertainty can introduce geometric drift between passes—part of what viewpoint-variation handling in the pipeline needs to correct for.

🎯 Planetary Geodetic Verification Framework

- Validation Baseline: Benchmarked against **USGS Lunar Control Networks & LRO LOLA DEMs**.
- Tested Regimes: Polar craters (Shackleton, Shoemaker) under extreme low-sun incidence (**>75°**).
- Outlier Tolerance: Maintains stability with up to **70%** feature occlusion in deep shadow basins.

GeoPhase-Lunar Pipeline



- **SPICE Photometric Inversion:**

Removes dynamic shadow bias using spacecraft orbital geometry.



- **Phase Congruency:** Illumination-invariant structural edges



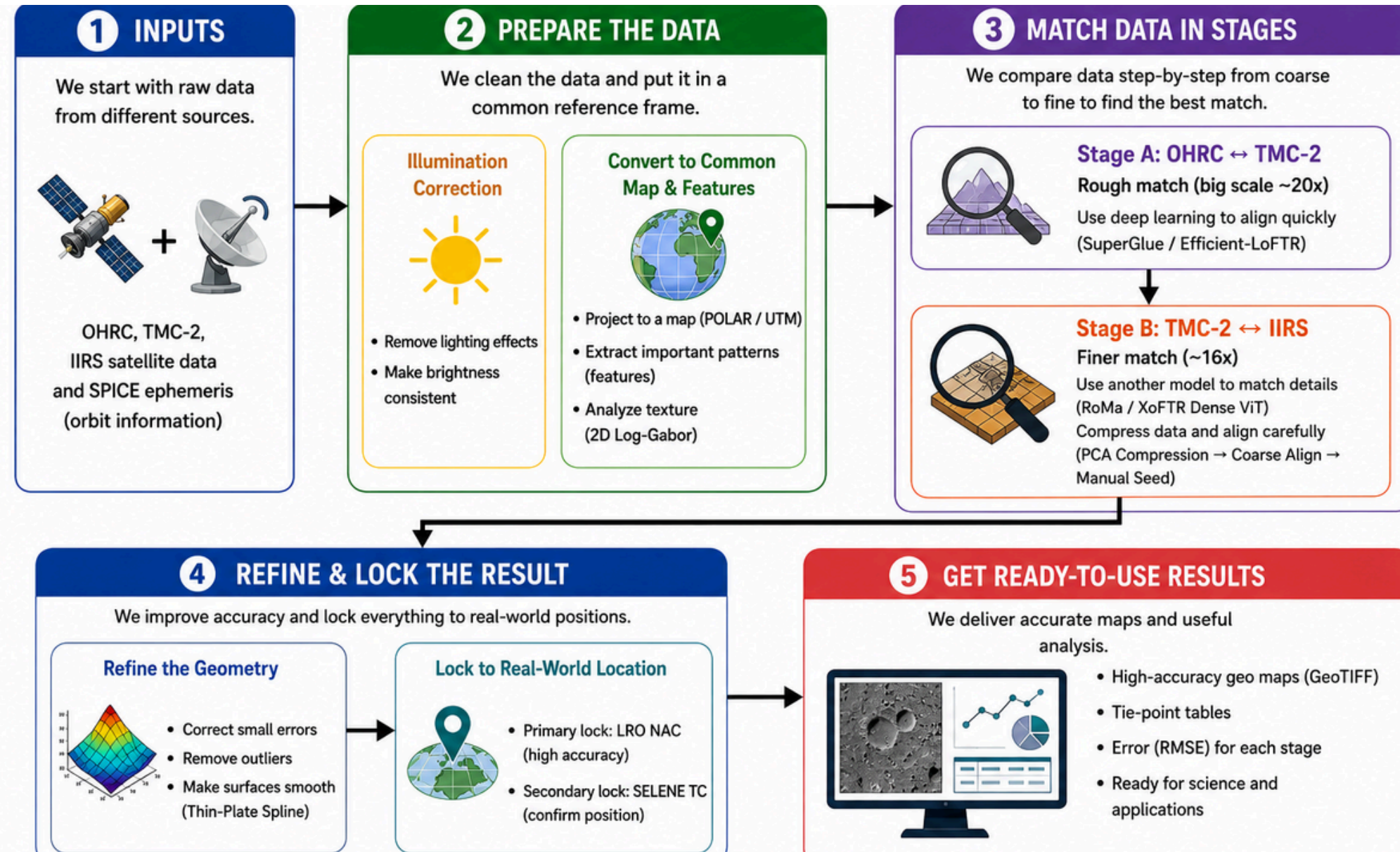
- **Grid NMS + MagSAC+++TPS:** Sub-pixel geodetic refinement



- **RoMa/XoFTR:** Dense cross-sensor matching

★ Why We Stand Out

- ✓ **Staged Resolution Pyramids:** Bridges the **320x** scale gap via an intermediate TMC-2 step, keeping each match within proven scale ranges.
- ✓ **Sub-Pixel Geodetic Precision:** Targets **RMSE < 1.0px** for optical stages and **< 2.0px** for cross-modal pairs.
- ✓ **Graceful Degradation:** Automatic fallback chain (PCA → coarse align → manual seed) points if the primary matcher underperforms.



- ❖ Core & Geospatial: Python 3.10+, GDAL, and Rasterio for PDS4 ingestion and georeferencing; SpiceyPy for orbital SPICE kernels and solar geometry.
- ❖ Deep Learning & Matching: PyTorch and ONNX Runtime powering SuperGlue / Efficient-LoFTR (optical) and RoMa / XoFTR (cross-spectral matching).
- ❖ Computer Vision & Refinement: OpenCV (MAGSAC++) for outlier rejection, with SciPy for DEM-guided Thin-Plate Spline (TPS) warping.
- ❖ Frequency Analysis: CuPy and NumPy for accelerated 2D Log-Gabor Phase Congruency structural extraction.
- ❖ UI & Deployment: Streamlit and FastAPI for the interactive ground station demo, packaged via Docker for zero-cloud local execution.

Technical Stack:

INGESTION



GEOSPATIAL



MATCHING



REFINEMENT

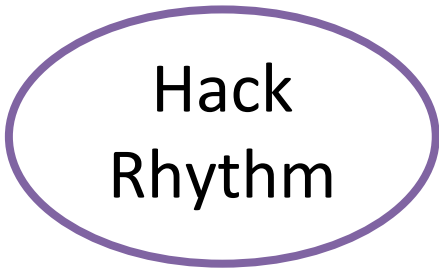


UI

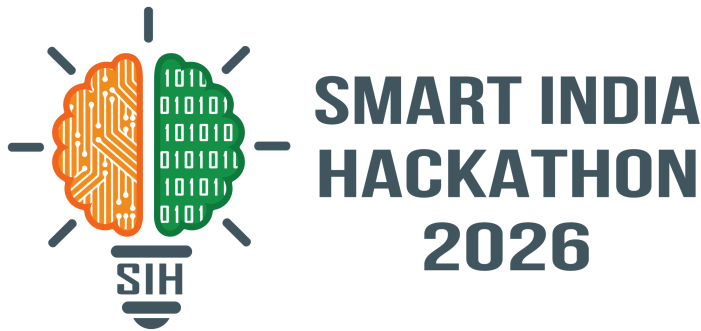


DEPLOY





FEASIBILITY AND VIABILITY



Feasibility:

The system is built to run entirely on standard laptops processing each image patch in under 1 second using less than 3.5 GB of GPU memory, with no cloud or server infrastructure required and integrates directly with official space data formats (PDS4/GeoTIFF) used by ISRO's PRADAN portal and NASA's LROC archive, grounding its results in optical physics rather than black-box AI.

Challenges / Risks	Strategy to Overcome
Extreme Shifting Shadows: at the lunar poles, low sun angles cast large moving shadows that trick normal computer vision into matching wrong shadow edges.	Physics-Based Shadow Removal (with Auto-Fallback): uses orbit metadata to mathematically remove shadows. If metadata is missing, it switches to contrast enhancement so it never fails.
Apples to Oranges: aligning a hyper-zoomed OHRC boulder photo with a low-res IIRS mineral map across a 320x zoom difference.	Stepping-Stone Strategy: align progressively instead of jumping the full 320x gap at once - High-Zoom (OHRC) -> Medium-Zoom (TMC-2) -> Hyperspectral (IIRS).
Rugged 3D Terrain: deep crater walls (>3 km) break normal flat 2D alignment and cause distortion/warping.	3D Elevation-Aware Alignment: uses lunar terrain height data to adjust for crater slopes, locking crater rims into place with sub-pixel precision.

Potential Impact on the Target Audience

- 🚀 **Reduces Manual Image Registration Effort (ISRO & Researchers):** Automates major stages of lunar image correspondence and alignment, reducing the time and effort required for manual tie-point selection and image preprocessing.
- 🌐 **Improves Lunar Mapping Under Difficult Conditions:** Enables more reliable matching across different Sun angles, resolutions, and sensor modalities, helping researchers analyse the same lunar region more consistently.
- 🛰️ **Supports Multi-Sensor Chandrayaan-2 Data Integration:** Helps bring OHRC, TMC-2, and IIRS data into a common correspondence framework, enabling combined geometric, visual, and spectral analysis.

Multi-Dimensional Benefits

- 💰 **Economic ROI & Cost Efficiency:** Saves millions in operational processing costs by creating an open, sovereign Indian mapping pipeline independent of expensive foreign commercial GIS software.
- 🇮🇳 **Strategic National Leadership:** Strengthens India's space software autonomy by providing a standardized, space-grade planetary mapping tool built for future lunar and planetary missions.
- 🌍 **Direct Dual-Use on Earth (Disaster Management & Defense):** The core AI transfers directly to Earth observation — matching all-weather SAR with optical images for rapid flood mapping, cyclone damage assessment, and defense surveillance.
- ⚡ **Future Clean Energy & Space Prospecting:** Provides the spatial precision needed to map critical lunar resources — water-ice deposits, regolith titanium, and clean-energy reserves — for the next era of space exploration.

Chandrayaan-2 & Lunar-Specific Studies

- [Analysis of Illumination Conditions in the Lunar South Polar Region Using Multi-Temporal High-Resolution Orbital Images](#) Remote Sensing 2023
- [Photometric correction of images obtained from Chandrayaan-2 IIRS data](#) Advances in Space Research, 2024
- [Lunar DEM and Orthoimage Generation from TMC-2 Stereo Data and Their Quality Assessment](#) Journal of the Indian Society of Remote Sensing 2023

Deep-Learning Feature Matching

- [LoFTR: Detector-Free Local Feature Matching with Transformers](#) CVPR, 2021
- [Efficient LoFTR: Semi-Dense Local Feature Matching with Sparse-Like Speed](#) CVPR, 2024
- [Self-Supervised Keypoint Detection and Cross-Fusion Matching Networks for Multimodal Remote Sensing Image Registration](#), Remote Sensing, vol. 14, no. 15, 2022

Multimodal & Illumination-Robust Matching

- [RIFT: Multi-Modal Image Matching Based on Radiation-Variation Insensitive Feature Transform](#) IEEE Trans. Image Processing, 2020
- [Phase Symmetry and Rank-Based Local Self-Similarity for Multimodal Remote-Sensing Registration](#) Int. J. Digital Earth, 2024

Surveys & Foundational Work

- [Multimodal Remote Sensing Image Registration Methods and Advancements: A Survey](#) Remote Sensing, 2021
- [Uniform Robust SIFT \(UR-SIFT\) for Optical Remote Sensing Images](#) IEEE Trans. Geoscience & Remote Sensing, 2011
- [Feature based remote sensing image registration techniques: A comprehensive and comparative review](#), Int. J. Remote Sensing, vol. 43, no. 12, 2022